

9 Light is essential for photosynthesis.

(a) Explain the role of light in the light-dependent reactions in photosynthesis.

(3)

- (b) The photograph shows thale cress, a small flowering plant found in Europe, Asia and Africa.



(Source: © blickwinkel / Alamy Stock Photo)

The effect of wavelength of light on thale cress was investigated.

Three groups of plants were treated with a different wavelength of light.

One group of plants was treated with blue light, one group with amber light and one group with red light.

The following were measured:

- rate of photosynthesis
- leaf biomass
- pigment content of the leaves
- changes in transcription of genes in the leaves.

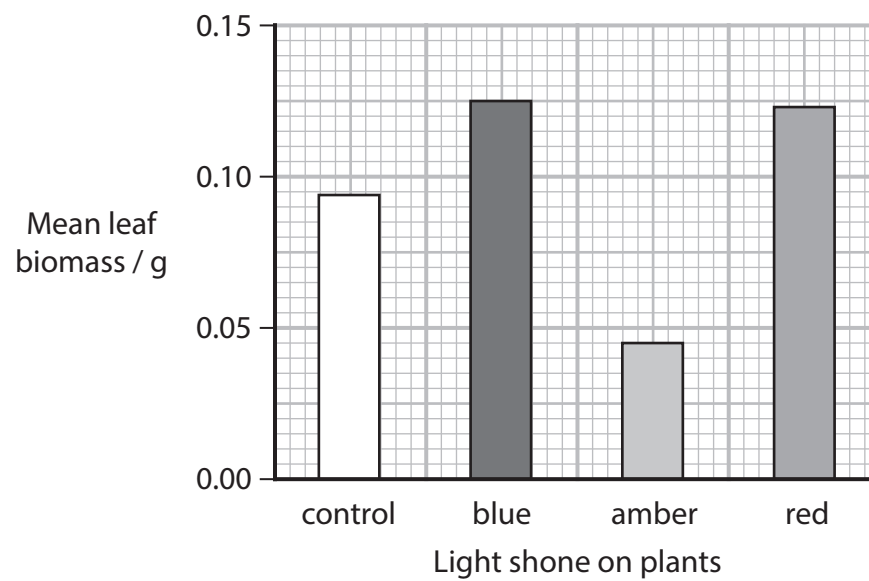
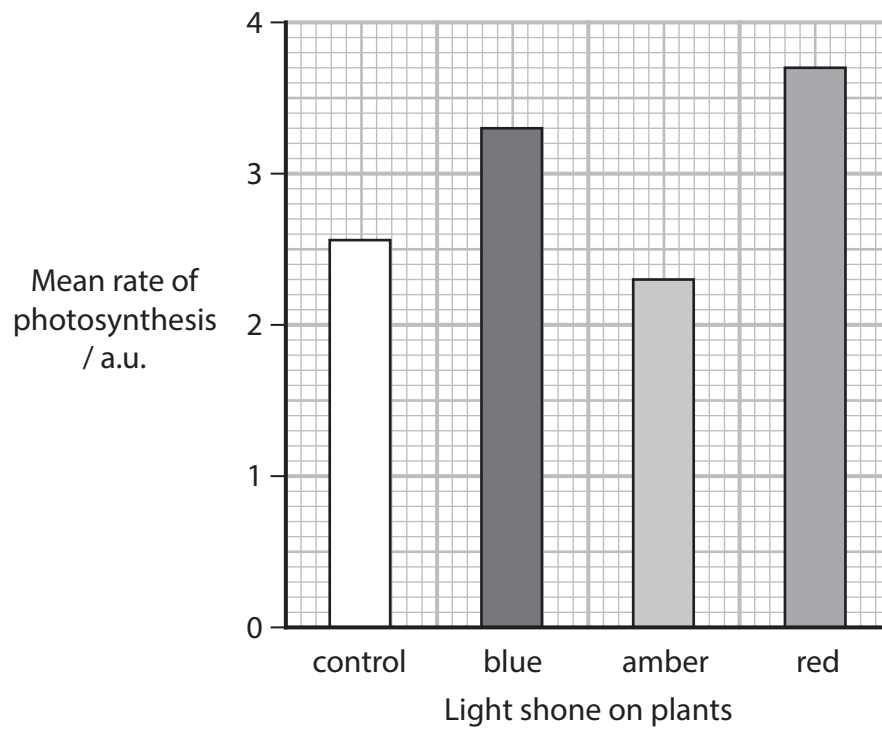
Another group of these plants had a control light shone on them.

- (i) Which would have been the most appropriate light to use as a positive control?

(1)

- ☐ **A** green
- ☐ **B** mixture of green, blue and red light
- ☐ **C** ultraviolet
- ☐ **D** sunlight

- (ii) The graphs show the results for the mean rate of photosynthesis and mean leaf biomass.



Explain the results shown in these graphs.

(4)

[illegible]

(iii) The table shows the results for the pigment content of the leaves.

Pigment	Mass of pigment / μg per g of leaf biomass			
	Control light	Blue light	Amber light	Red light
Chlorophyll a	382	412	315	441
Chlorophyll b	148	148	125	133
Anthocyanins	78	79	71	72

Describe **three** conclusions that can be made from these results.

(3)

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- (iv) Plants treated with blue light had an increase in photosynthesis caused by an increase in the transcription of some genes in the leaves.

Suggest **two** genes whose transcription could be increased in the presence of blue light.

Give a reason for suggesting each of these genes.

(3)

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(Total for Question 9 = 14 marks)

TOTAL FOR PAPER = 90 MARKS